TABLEAU · LESSON 2 OF 6

Charts in Tableau: From Bar and Line to Maps, Dual Axis, Histograms and Pareto

Build every core chart in Tableau and know when to use each: bar, stacked and side-by-side bars, line and area, pie and donut, treemap, highlight table, scatter plot, histogram with bins, box plot, filled and symbol maps, dual-axis combo charts, bullet charts and a Pareto chart, all on the Superstore India data.

✦ Tableau Desktop / Tableau Public 2026 | Beginner | ⏱ 150 min

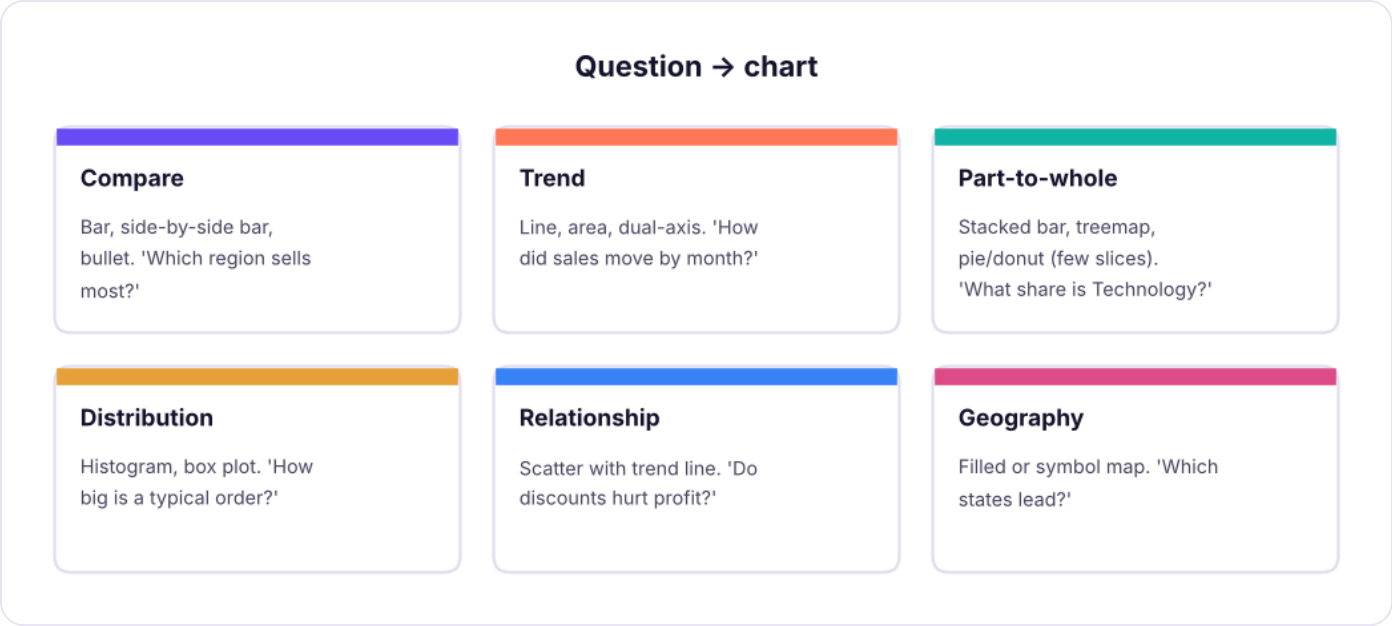
By the end of this lesson you can

- Choose the right chart for comparison, trend, part-to-whole, distribution, relationship and geography questions
- Build bar, line, area, pie, donut, treemap and highlight-table views
- Create bins and histograms, box plots and scatter plots with trend lines
- Map states with geographic roles, and combine two measures with dual and blended axes
- Build a bullet chart and a Pareto (80/20) chart

Datasets used

`superstore_sales.csv`

Choose the chart from the question



Match the business question to the chart family.

All examples below use the Superstore India data. The tables show the numbers your Tableau view should display, computed here with pandas so you can check your work.

Python

```
import pandas as pd, numpy as np
df = pd.read_csv("superstore_sales.csv", parse_dates=["order_date"])
df["year"] = df.order_date.dt.year
print(len(df), "order lines,", df.order_id.nunique(), "orders,", df.customer_id.nunique(),
"customers")
```

OUTPUT

```
6323 order lines, 3200 orders, 785 customers
```

Bar charts

Variant	Build	Use
Bar	Dimension on Rows, measure on Columns	Ranking categories
Stacked bar	Add a second dimension to **Color**	Total and composition
Side-by-side bar	Second dimension on Columns *before* the measure (or Show Me → side-by-side)	Comparing subgroups directly
100% stacked	Stacked bar + Quick Table Calculation → **Percent of Total** (compute using the colour dimension)	Mix comparisons

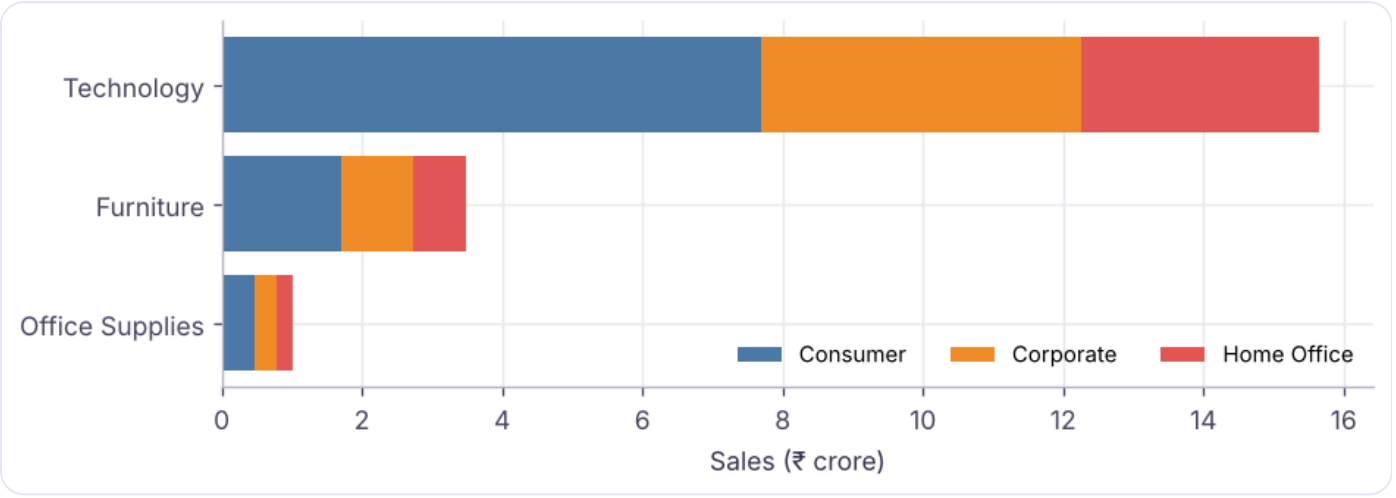
Python

```
seg = pd.crosstab(df.category, df.segment, values=df.sales, aggfunc="sum") / 1e7
seg.round(2)
```

OUTPUT

segment	Consumer	Corporate	Home Office
category			
Furniture	1.70	1.03	0.75
Office Supplies	0.47	0.32	0.23
Technology	7.69	4.56	3.39

3 rows × 3 columns



Stacked bar: Sales by Category, coloured by Segment (₹ crore).

Line and area charts

Put a **continuous** date (green MONTH(Order Date)) on Columns and Sales on Rows for one timeline. Put **YEAR** on Color and a **discrete** MONTH on Columns to compare the same months across years (seasonality).

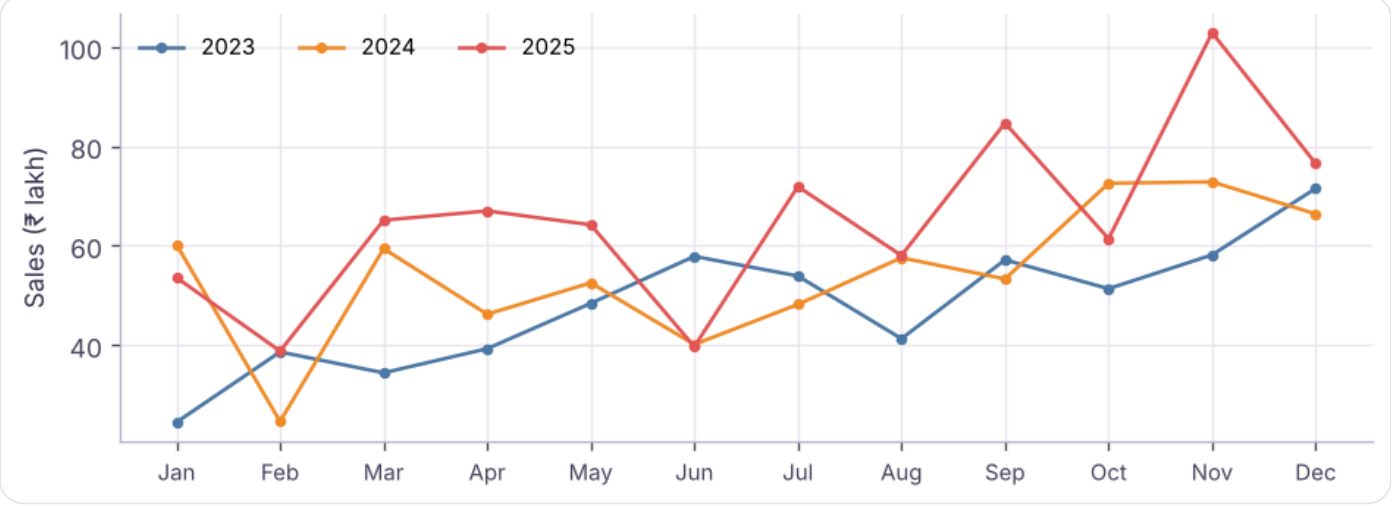
Python

```
pv = df.pivot_table(index=df.order_date.dt.month, columns="year", values="sales",
aggfunc="sum") / 1e5
pv.index.name = "month"
pv.round(1).head(6)
```

OUTPUT

year	2023	2024	2025
month			
1	24.5	60.2	53.6
2	38.6	24.6	38.8
3	34.4	59.4	65.1
4	39.3	46.2	67.0
5	48.4	52.5	64.2
6	57.9	40.1	39.7

6 rows × 3 columns



Discrete month on Columns, Year on Color: each year is a line, so seasonality is easy to compare.

An **area chart** is the same view with Mark type = Area; stack a dimension on Color to show how the total is built up.

Part-to-whole: pie, donut, treemap

- **Pie**: Mark type Pie, dimension on Color, measure on **Angle**. Use for 2–5 slices only.
- **Donut**: two pies on a dual axis. Put `AVG(0)` (type `avg(0)` on Rows) twice, make the second smaller and white, turn on dual axis, and put the total on the inner label.
- **Treemap**: dimension on **Detail/Label**, measure on **Size** and **Color**. Good for many categories (e.g. the 13 sub-categories).

Python

```
sub = (df.groupby("sub_category").agg(sales=("sales", "sum"), profit=("profit", "sum")) /
1e5).sort_values("sales", ascending=False)
sub["margin %"] = (sub.profit / sub.sales * 100).round(1)
sub.round(1).head(8)
```

OUTPUT

	sales	profit	margin %
sub_category			
Copiers	787.5	84.4	10.7
Machines	425.9	51.9	12.2
Phones	319.2	36.6	11.5
Tables	140.5	-8.2	-5.8
Bookcases	112.7	-4.2	-3.7
Chairs	80.9	3.4	4.2
Appliances	63.3	9.4	14.9
Accessories	31.8	3.7	11.6

8 rows × 3 columns

Highlight table (heat map)

Category on Rows, YEAR on Columns, Sales on **Text** and **Color**, Mark type **Square**. Tableau shades each cell, so high and low cells stand out without reading every number.

Distribution: histogram and box plot

STEP BY STEP

Histogram of order value

1. Right-click **Sales** → Create → **Bins**; set the bin size (e.g. ₹10,000). A new dimension "Sales (bin)" appears.
2. Drag **Sales (bin)** to Columns and **CNT(Sales)** (or Number of Records) to Rows.
3. Or select Sales and pick the histogram in Show Me: it creates the bins for you.

Python

```
bins = pd.cut(df.sales, bins=[0, 5000, 10000, 20000, 50000, 100000, df.sales.max()],
right=False)
print(bins.value_counts().sort_index())
print("median line value: ₹", round(df.sales.median()), "| mean: ₹",
round(df.sales.mean()))
```

OUTPUT

```
sales
[0.0, 5000.0)           2530
[5000.0, 10000.0)        829
[10000.0, 20000.0)       896
[20000.0, 50000.0)       942
[50000.0, 100000.0)      546
[100000.0, 619290.0)     579
Name: count, dtype: int64
median line value: ₹ 8430 | mean: ₹ 31854
```

The mean is well above the median: a few large orders pull it up (right skew). A **box plot** (Show Me → box-and-whisker) shows the median, quartiles and outliers per category at a glance.

Relationship: scatter plot

Put SUM(Sales) on Columns, SUM(Profit) on Rows and **Sub-Category** on Detail: one dot per sub-category. Add **Analytics pane → Trend Line** (linear) and hover it to read R^2 and the p-value.

Python

```
s = df.groupby("sub_category")[["sales", "profit", "discount"]].agg({"sales": "sum",
"profit": "sum", "discount": "mean"})
print("correlation sales vs profit across sub-categories:", round(s.sales.corr(s.profit),
2))
print("correlation avg discount vs profit margin:", round(s.discount.corr(s.profit /
s.sales), 2))
```

OUTPUT

```
correlation sales vs profit across sub-categories: 0.95
correlation avg discount vs profit margin: -0.88
```

Maps

1. Make sure **State** has the geographic role *State/Province* (and the country is India: Map → Edit Locations).
2. Double-click State: Tableau adds Latitude/Longitude (generated) and draws a **symbol map**. Change Mark type to **Map** for a **filled (choropleth) map**.
3. Put Sales on Color for a filled map, or on Size for a symbol map. Use Profit on Color with a diverging palette (orange-blue) to show loss-making states.

Python

```
st = (df.groupby("state")[["sales", "profit"]].sum() / 1e5).round(1).sort_values("sales",
ascending=False)
st.head(6)
```

OUTPUT

	sales	profit
state		
Maharashtra	256.7	24.4
Delhi	156.6	17.3
Kerala	147.1	12.6
Madhya Pradesh	144.1	12.7
West Bengal	138.9	14.6
Karnataka	136.3	11.8

6 rows × 2 columns

> **TIP** **Indian maps**
>
> Tableau's built-in geocoding knows Indian states and many districts and cities. If a name does not match (old spellings, "Orissa" vs "Odisha"), use **Map → Edit Locations** to match it, or join a custom latitude/longitude file.

Dual-axis and combo charts

Show two measures with different scales, e.g. Sales as bars and Profit margin as a line:

1. Continuous MONTH(Order Date) on Columns; SUM(Sales) and the margin calculation on Rows.
2. Right-click the second pill → **Dual Axis**. Set the Marks card for Sales to **Bar** and for margin to **Line**.
3. Right-click the right axis → **Synchronize Axis** only when both measures share units (Sales and Profit); never synchronise ₹ with %.

A **blended axis** (drag a second measure onto the *existing* axis until the double-ruler icon shows) puts two measures with the same unit on one axis, using Measure Names on Color.

Bullet chart (actual vs target)

Select a measure and a target (e.g. Sales and a target field) → Show Me → **Bullet graph**. The bar is the actual, the reference line is the target, and the shaded bands are % of target (60%, 80%). If the bars and lines come out the wrong way round, right-click the axis → **Swap Reference Line Fields**.

Pareto chart (80/20)

> **STEP BY STEP** **Build a Pareto of products**
>
> 1. Product Name on Columns sorted by SUM(Sales) descending; SUM(Sales) on Rows as bars.
> 2. Drag SUM(Sales) to Rows again → Quick Table Calculation → **Running Total**, then Edit Table Calculation → *Add secondary calculation* → **Percent of Total**.
> 3. Make it dual axis; set the second mark to Line; add a reference line at 80%.
> 4. Optional: replace Product Name on Columns with a table calc `INDEX()/SIZE()` so the x-axis reads "% of products".

Python

```python
p = df.groupby("product_name").sales.sum().sort_values(ascending=False)
cum = p.cumsum() / p.sum()
n80 = int((cum < .8).sum()) + 1
print(f"{n80} of {len(p)} products ({n80 / len(p):.0%}) make 80% of sales")
```

OUTPUT

```
21 of 78 products (27%) make 80% of sales
```

Sales are concentrated: about a quarter of the products bring in 80% of sales, close to the classic 80/20 rule. Those top products deserve the best stock availability and marketing. A Pareto chart shows how concentrated a business really is, so you can judge whether a "focus on the top products" strategy makes sense.

Summary

KEY TAKEAWAYS

- Pick the chart from the question: compare, trend, part-to-whole, distribution, relationship, geography.
- Continuous dates draw timelines; discrete dates compare periods.
- Bins create histograms; the Analytics pane adds trend lines, reference lines and box plots.
- Assign geographic roles before mapping; use filled maps for rates and symbol maps for totals.
- Dual axis combines two marks (bar + line); synchronise only when units match.

Quick quiz

6 questions · tap an answer to check it

1. Which chart best shows how a single order-value variable is distributed?

- **A** Histogram
- **B** Pie chart
- **C** Treemap
- **D** Line chart

Answer: A. Histograms show distributions; create bins first.

2. How do you get Profit as a line on top of Sales bars with its own scale?

- **A** Put both on Rows, use Dual Axis, set one mark to Bar and the other to Line
- **B** Put Profit on Color
- **C** Use a blended axis
- **D** Synchronise the axes

Answer: A. Dual axis gives two independent axes and separate mark types.

3. Which is the right way to show 13 sub-categories as a part-to-whole?

- **A** Treemap or sorted bar chart
- **B** Pie chart with 13 slices
- **C** Scatter plot
- **D** Line chart

Answer: A. Pies stop being readable beyond about five slices.

4. You want each year as its own line to compare seasonality. What goes on Columns?

- **A** Discrete MONTH(Order Date)
- **B** Continuous MONTH(Order Date)
- **C** YEAR(Order Date) only
- **D** Sales

Answer: A. Discrete month gives Jan–Dec headers; Year goes on Color.

5. What does the Pareto line show?

- **A** The running total of sales as a percent of the grand total
- **B** Profit margin
- **C** Average sales
- **D** The number of orders

Answer: A. Running total + percent of total secondary calculation.

6. When should you synchronise a dual axis?

- **A** When both measures share the same unit and scale makes sense
- **B** Always
- **C** When one is in ₹ and the other in %
- **D** Never

Answer: A. Synchronising ₹ with % distorts the chart.

EXERCISE **Practice**

Build: (1) a side-by-side bar of Sales by Region and Segment; (2) a filled map of Profit by State with a diverging palette; (3) a histogram of Sales with ₹10,000 bins; (4) a dual-axis chart of monthly Sales (bars) and Profit (line); (5) a Pareto chart of sub-categories with an 80% reference line.

Show solution (worked answer)

Use the steps in each section. For the map, set Profit on Color → Edit Colors → Orange-Blue Diverging, centre 0. For the Pareto use Running Total + Percent of Total, compute using Sub-Category, then Analytics → Reference Line → Constant 0.8 on the line axis.

INTERVIEW CORNER **Questions you may be asked**

Q: Dual axis vs blended axis? A dual axis gives two independent axes (left and right) and allows different mark types; a blended axis puts several measures on one shared axis (same unit) using Measure Names/Values.

Q: How do you build a Pareto chart in Tableau? Sort the dimension by the measure, show bars, add the measure again with a Running Total table calculation and a secondary Percent of Total, make it dual axis as a line, and add an 80% reference line.

Q: Filled map or symbol map? Filled maps suit rates and ratios (profit margin by state) because area does not scale with value; symbol maps with size suit totals (sales by city).

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